IBM FileNet P8 V5.6.x

Step-by-Step Hands-On Lab Guide

Day 8 - IBM Content Navigator (ICN)

Installation | Architecture | Core Features | Customization | Labs

Author	Course Day	Version
Rajesh Pasham	Day 8 of 10	ICN 3.0.11

Lab Overview

Item	Description
Lab Title	IBM Content Navigator - Installation, Configuration & Lab Exercises
Course Day	Day 8: IBM Content Navigator (ICN)
Author	Rajesh Pasham
Lab Duration	Approximately 3 Hours (Installation 45 min + Configuration Labs 135 min)
Environment	Content Platform Engine installed, P8Domain configured, TrainingOS Object Store created
Deliverables	Fully configured ICN 3.0.11 with Training Desktop, Repository Connection, Search Templates, and Teamspace

> **START STATE:** Content Platform Engine 5.2.1 is installed. P8Domain exists. TrainingOS Object Store is configured. ICN is NOT yet installed. You will install it in Lab Exercise 1.

LAB EXERCISE 1 - ICN Installation

Install IBM Content Navigator 3.0.11 on WebSphere Application Server

Attribute	Details
Exercise	LAB-08-01
Duration	45 minutes
Difficulty	Intermediate - requires WAS and DB2 familiarity

STEP 1 Create ICNDB Database in DB2

Open a Command Prompt as Administrator and run the following DB2 commands:

1. Open Start Menu > IBM DB2 > DB2 Command Window (Administrator)
2. Create the ICN configuration database:

```
db2 CREATE DATABASE ICNDB AUTOMATIC STORAGE YES USING CODESET UTF-8 TERRITORY US COLLATE
USING SYSTEM PAGESIZE 32768
```

3. Connect and grant permissions:

```
db2 CONNECT TO ICNDB
db2 GRANT DBADM ON DATABASE ICNDB TO USER ceadmin
db2 DISCONNECT ICNDB
```

4. Verify the database was created:

```
db2 LIST DB DIRECTORY
```

	Verification Check	Expected Result
[ ]	ICNDB appears in the database directory listing	*Status: Indirect*
[ ]	Codeset shows UTF-8	*UTF-8 encoding confirmed*
[ ]	No errors during GRANT command	*Command completed successfully*

STEP 2 Configure ECMClientDS Data Source in WebSphere

5. Open WAS Admin Console: https://localhost:9043/ibm/console
6. Navigate to: Environment > Shared Libraries > New
7. Create a shared library named DB2_JDBC with these JAR file paths:

```
C:\IBM\SQLLIB\java\db2jcc4.jar
C:\IBM\SQLLIB\java\db2jcc_license_cu.jar
```

8. Click Apply and Save
9. Navigate to: Resources > JDBC > JDBC Providers > New
10. Create a DB2 JDBC provider using: DB2 Using IBM JCC Driver (XA)
11. Navigate to: Resources > JDBC > Data Sources > New
12. Fill in these exact values:

Field	Value	Notes
Data Source Name	**ICN Data Source**	*Descriptive name only*
JNDI Name	**ECMClientDS**	*MUST be exactly this value*
Database Name	**ICNDB**	*The database created in Step 1*
Server Name	**localhost**	*DB2 server host*
Port	**50000**	*DB2 listener port*
Username	**ceadmin**	*Has DBADM on ICNDB*

13. Save the data source, then click Test Connection

	Verification Check	Expected Result
[ ]	JNDI name ECMClientDS entered correctly	*Exact match - no typos*
[ ]	Connection test succeeds	*Message: The test connection operation was successful*
[ ]	DB2 JDBC JARs visible in shared library	*db2jcc4.jar listed*

STEP 3 Install ICN via IBM Installation Manager

14. Launch IBM Installation Manager as Administrator
15. Click File > Preferences > Repositories > Add Repository
16. Browse to: <software_repository>\ICN_3.0.11_WIN_ML\repository.config
17. Click Install from the IM main screen
18. Select IBM Content Navigator 3.0.11 and accept the license
19. Set installation directory: C:\IBM\ContentNavigator
20. On the Configuration page, select WAS profile directory:

```
C:\IBM\WebSphere\AppServer\profiles\FileNetServer
```

21. Click Next, review the installation summary, then click Install
22. Monitor the IM log - wait for Installation completed successfully message

[Screenshot Placeholder]

Figure: *IBM Installation Manager - ICN 3.0.11 installation progress showing navigator.war deployment*

	Verification Check	Expected Result
[ ]	IM log shows: Installation completed successfully	*No errors in IM log*

[ ]	ICN installed to C:\IBM\ContentNavigator	*Directory exists with ICN files*
[ ]	navigator.war present in WAS deployment directory	*Check WAS admin console > Applications*

STEP 4 Run ICN Configuration Tool and Verify Installation

23. Restart WAS: Open Services (services.msc) > IBM WebSphere Application Server > Restart
24. Wait 2-3 minutes for WAS to fully restart
25. Open browser and navigate to the ICN Configuration Tool:

```
http://localhost:9080/navigator/config
```

26. Click Run Configuration Tool
27. On Data Source Configuration page, verify:
    - JNDI Name: ECMClientDS (should be auto-populated)
    - Schema Owner: ceadmin
    - Action: Initialize new configuration database
28. On Administrator Setup page, enter ceadmin credentials
29. Click Run - watch the progress log until: Configuration completed successfully
30. Click Finish, then restart WAS one final time
31. After WAS restart, navigate to ICN:

```
http://localhost:9080/navigator
```

32. Log in with ceadmin - ICN Admin Desktop should load

[Screenshot Placeholder]

Figure: *ICN Admin Desktop - left panel showing Administration section with Repositories, Desktops, Plug-ins nodes*

	Verification Check	Expected Result
[ ]	ICN login page loads at http://localhost:9080/navigator	*Login screen displayed*
[ ]	Login with ceadmin succeeds	*Admin Desktop loads with Administration panel*
[ ]	Configuration Tool ran successfully	*Message: Configuration completed successfully*
[ ]	ICNDB tables created in DB2	*Run: db2 list tables for schema ICNUSER*

LAB EXERCISE 2 - Repository Connection & Desktop Configuration

Connect ICN to FileNet CE and create the Training Desktop

Attribute	Details
Exercise	LAB-08-02
Duration	30 minutes
Prerequisite	LAB-08-01 completed - ICN installed and running

STEP 1 Create FileNet Repository Connection

33. Open ICN Admin Desktop: http://localhost:9080/navigator/?desktop=admin
34. In left panel: Administration > Repositories > New
35. Fill in the New Repository form:

Field	Value	Notes
Repository Type	**IBM FileNet Content Manager**	*Select from the type dropdown*
Repository ID	**FNCE_TrainingOS**	*No spaces allowed*
Display Name	**FileNet Training Repository**	*Shown to users in desktop picker*
Server URL	**http://localhost:9080/wsi/FNCEWS40MTOM/**	*CE Web Services endpoint*
JAAS Login	**FileNetP8**	*Default JAAS login for FileNet P8*
User Name	**ceadmin**	*CE administrator account*
Password	**<ceadmin password>**	*Your CE admin password*

36. Click Connect - the Object Store list should populate
37. Select TrainingOS from the Object Store list
38. Click the Workflow tab and configure:
    - Enable Workflow: Checked
    - Connection Point: PE_ConnectionPoint
39. Click Save

[Screenshot Placeholder]

Figure: *ICN Admin Desktop - New Repository dialog with FileNet CE connection details filled in and Object Store list populated*

	Verification Check	Expected Result
[ ]	Connect test succeeds and Object Store list populates	*TrainingOS visible in list*
[ ]	Repository saved as FNCE_TrainingOS	*Appears in Repositories list*
[ ]	Workflow enabled with PE_ConnectionPoint	*Workflow tab configured*

STEP 2 Create the Training Desktop

40. In Admin Desktop: Administration > Desktops > New
41. General tab - fill in:

Field	Value	Notes
Desktop ID	**TrainingDesktop**	*Used in URL parameter ?desktop=TrainingDesktop*
Display Name	**FileNet Training Desktop**	*Shown on login screen desktop picker*
Description	**Desktop for IBM FileNet P8 Day 8 Labs**	
Set as Default	**Checked**	*Makes this the default desktop for all users*

42. Layout tab: Select Two-pane layout. Default Feature: Browse. Default Action: View.
43. Repositories tab: Click Add Repository > select FileNet Training Repository > Add
44. Features tab: Enable all of these:
    - Browse - checked
    - Search - checked
    - Teamspaces - checked
    - Work - checked
    - Favorites - checked
45. Click Save to save the Training Desktop
46. Test the Training Desktop:

```
http://localhost:9080/navigator/?desktop=TrainingDesktop
```

[Screenshot Placeholder]

Figure: *Training Desktop loaded in browser - showing Browse, Search, Teamspaces, Work, and Favorites feature tabs*

	Verification Check	Expected Result
[ ]	TrainingDesktop saved and appears in Desktops list	*Entry visible in Admin Desktop*

[ ]	Training Desktop loads at ?desktop=TrainingDesktop	*ICN UI loads with Training Desktop layout*
[ ]	All 5 feature tabs visible in Training Desktop	*Browse, Search, Teamspaces, Work, Favorites*
[ ]	TrainingOS Object Store accessible via Browse	*Folder tree shows TrainingOS structure*

LAB EXERCISE 3 - Create Search Templates

Build metadata and full-text search templates for document retrieval

Attribute	Details
Exercise	LAB-08-03
Duration	30 minutes
Prerequisite	LAB-08-02 completed - Training Desktop configured with TrainingOS repository

STEP 1 Create a Metadata Search Template

You will create a search template that allows users to search for documents by Document Title and Document Date. This is a metadata-only search - no full-text indexing required.

47. In Admin Desktop: Administration > Searches > New Search Template
48. General tab - fill in:

Field	Value	Notes
Name	**Document Search by Title**	*Display name shown to users*
Description	**Search documents by title and date range**	
Repository	**FileNet Training Repository**	*Select from dropdown*
Object Store	**TrainingOS**	
Document Classes	**Document (base class)**	*Searches all document types*

49. Search Criteria tab - click Add Property to add search fields:

CE Property	Operator	Configuration
Document Title	Contains	Required: No, Label: Document Title
Date Created	Between	Required: No, Label: Created Between
Creator	Equals	Required: No, Label: Created By

50. For Document Title: set Display Name to 'Document Title', operator Contains, not required
51. For Date Created: set Display Name to 'Created Between', operator Between, not required
52. For Creator: set Display Name to 'Created By', operator Equals, not required
53. Display Properties tab - select these columns for search results:
    - Document Title
    - Date Created
    - Creator
    - Class Name
    - Document Number (if available)
54. Sort tab: Default sort by Date Created, Descending

55. Click Save

[Screenshot Placeholder]

Figure: *Search Template editor - Criteria tab showing Document Title (Contains), Date Created (Between) fields configured*

	Verification Check	Expected Result
[ ]	Search template saved as: Document Search by Title	*Appears in Searches list*
[ ]	Three search criteria fields configured	*Title, Date, Creator visible*
[ ]	Display properties set for results grid	*5 columns configured*

STEP 2 Add Search Template to Training Desktop

56. In Admin Desktop: Administration > Desktops > TrainingDesktop > Edit
57. Click the Search tab in the Desktop editor
58. Under Available Searches, find Document Search by Title
59. Click Add to move it to Selected Searches for this desktop
60. Click Save to update the Training Desktop
61. Open the Training Desktop in a new tab:

```
http://localhost:9080/navigator/?desktop=TrainingDesktop
```

62. Click the Search feature tab in the Training Desktop
63. Confirm that Document Search by Title appears in the search template list
64. Click the search template to open the search form
65. Enter a partial document title or leave blank and click Search

[Screenshot Placeholder]

Figure: *Training Desktop - Search tab showing Document Search by Title template selected and search form with criteria fields*

	Verification Check	Expected Result
[ ]	Search template appears in Training Desktop Search tab	*Template visible in search list*
[ ]	Search form opens with correct criteria fields	*Title, Date, Creator fields shown*
[ ]	Executing the search returns results from TrainingOS	*Documents from TrainingOS listed*

STEP 3 Create a Full-Text Search Template (if IFCSS is enabled)

> **PREREQUISITE:** This step requires IFCSS (Content Search Services 5.5.8) installed and full-text indexing enabled on TrainingOS in ACCE. If IFCSS is not installed in your lab, skip to Exercise 4.

66. In Admin Desktop: Administration > Searches > New Search Template
67. General tab: Name: Full-Text Document Search, Repository: FileNet Training Repository
68. Search Criteria tab: Click Add Text Search Criteria
69. Configure the full-text search field:
    - Label: Search Text (All Words)
    - Type: Full Text
    - Search Mode: All Words (AND mode)
70. Add a second criterion: Document Title Contains (metadata filter to narrow results)
71. Display Properties: add Document Title, Relevance Score, Date Created, Creator
72. Sort by: Relevance Score, Descending
73. Click Save, then add to Training Desktop Search tab
74. Test: type a word that appears in a document's content and run the search

	Verification Check	Expected Result
[ ]	Full-text search template created and saved	*Appears in Searches list*
[ ]	Template added to Training Desktop	*Visible in Search tab*
[ ]	Full-text search returns documents containing the search term	*Results include relevance score column*

LAB EXERCISE 4 - Configure ICN Desktop

Configure menus, toolbars, and create a teamspace in the Training Desktop

Attribute	Details
Exercise	LAB-08-04
Duration	35 minutes
Prerequisite	LAB-08-02 and LAB-08-03 completed

STEP 1 Create a Custom Document Context Menu

75. In Admin Desktop: Administration > Menus > New
76. Fill in: Menu ID: TrainingDocMenu, Display Name: Training Document Menu
77. From Available Actions panel, add these actions to Selected Actions (in order):
    - View Document
    - Open Document
    - Separator (---)
    - Check Out
    - Check In
    - Separator (---)
    - View Properties
    - View Versions
    - View Audit Trail
    - Separator (---)
    - Add to Favorites
    - Copy
    - Move
    - Separator (---)
    - Delete
78. Click Save
79. Go to Administration > Desktops > TrainingDesktop > Edit
80. Click the Menus tab
81. For Document Context Menu: change from Default to TrainingDocMenu
82. Click Save

	Verification Check	Expected Result
[ ]	Training Document Menu created with 13 actions and 4 separators	*Saved in Menus list*
[ ]	Training Desktop uses TrainingDocMenu for document context	*Menus tab shows custom assignment*
[ ]	Right-clicking a document shows the new custom menu	*Correct actions listed in menu*

STEP 2 Create a Teamspace

Now create a teamspace from the Training Desktop to verify the Teamspaces feature is working correctly. This simulates a real project workspace creation.

83. Open the Training Desktop: http://localhost:9080/navigator/?desktop=TrainingDesktop
84. Log in as ceadmin
85. Click the Teamspaces tab in the left feature panel
86. Click New Teamspace (or the + button in the toolbar)
87. Fill in the teamspace details:

Field	Value	Notes
Teamspace Name	**Training Project Alpha**	*Descriptive project name*
Description	**Collaborative workspace for FileNet P8 training exercises**	
Repository	**FileNet Training Repository**	*TrainingOS Object Store*
Template	**No Template (create blank)**	*Or use a template if configured*

88. Click Next to configure the teamspace folder structure:
    - Add folder: Documents
    - Add folder: Reviews
    - Add folder: Approvals
    - Add folder: Archives
89. Click Next to configure membership:
    - Add ceadmin as Owner
90. Click Finish to create the teamspace
91. Verify the teamspace appears in the Teamspaces tab
92. Click into the teamspace and confirm the 4 subfolders are visible
93. Try adding a document to the Documents folder within the teamspace

[Screenshot Placeholder]

Figure: *Training Project Alpha teamspace open in ICN - showing Documents, Reviews, Approvals, Archives subfolders*

	Verification Check	Expected Result
[ ]	Training Project Alpha teamspace created successfully	*Appears in Teamspaces tab*
[ ]	4 subfolders created inside the teamspace	*Documents, Reviews, Approvals, Archives visible*
[ ]	ceadmin is the teamspace Owner	*Owner badge shown next to ceadmin*
[ ]	Document can be added to teamspace Documents folder	*Upload succeeds, doc appears in list*

STEP 3 Use Favorites Feature

94. In the Training Desktop, click the Browse tab and navigate to any folder in TrainingOS
95. Right-click a document in the content list
96. Select Add to Favorites from the context menu
97. Click the Favorites tab in the left panel
98. Confirm the document appears as a favorite
99. Click the favorite to verify it navigates directly to the document
100. Right-click the favorite and explore: Remove from Favorites option

	Verification Check	Expected Result
[ ]	Document added to Favorites successfully	*Appears in Favorites tab with name and repository*
[ ]	Clicking the favorite opens the document	*Document viewer or properties opens*
[ ]	Favorites persist after browser refresh	*Still visible after pressing F5*

Lab Completion Summary

Lab	Topic	Completion State
LAB-08-01	ICN Installation	ICN 3.0.11 installed, ICNDB initialized, ICN accessible at /navigator
LAB-08-02	Repository & Desktop	FileNet repository connected, Training Desktop with all 5 features configured
LAB-08-03	Search Templates	Metadata search + full-text search templates created and tested
LAB-08-04	Desktop Config	Custom menus, teamspace, and favorites fully configured and verified

> **NEXT:** Day 9 - API Integration & Automation. You will use the CMIS REST API to upload documents and trigger workflow actions on the TrainingOS repository you configured today.